

EVALUATION IN NLP

Types of Evaluations

- **Intrinsic** - How well does the algorithm do at its task?
- **Extrinsic** - How well does it facilitate a larger task of which it is a component?
- **Situated** - How well does it perform in a real-use setting?
- **Economic** - What will people pay for it?

Example - Google PageRank

- Intrinsic
 - Controlled set of queries and documents and relevance assessments
 - Formally defined evaluation measures - precision, recall, runtime, etc.
- Extrinsic
 - External task, e.g. helping people answer a “factoid” question
 - Formal measures for that external task, e.g. user time to answer questions
 - Controlled experiment, but interaction with external factors situated
 - In-use study (monitoring real searches with their questions)
 - Hard to conduct a controlled, replaceable experiment

What Is an Economic Evaluation?

- People “vote with their pocketbooks”
- In addition to that, one can add subjective surveys, etc., instrumentation
- Many uncontrolled variables, sales/marketing manipulation, etc.

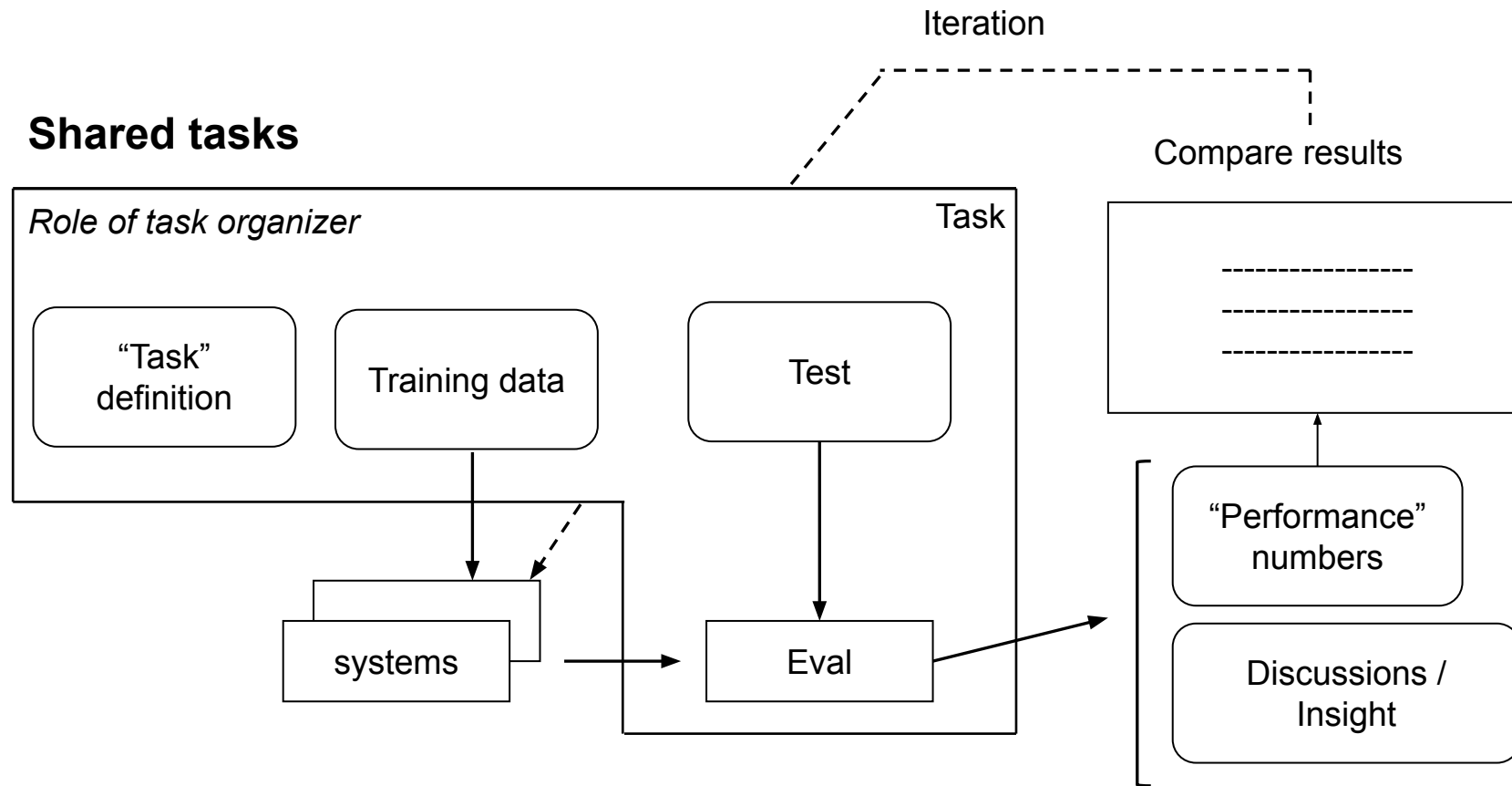
Formative vs. Summative

- **Formative** informs algorithm designer about progress (fast, iterative)
 - When the cook tastes the soup
- **Summative** assesses whether goals are met at a major milestone
 - When the customer tastes the soup
- Historically, NLP / AI evaluations required a lot of manual effort
- Breakthrough comes with the implementation of automatic evaluations (BLEU, PARSEVAL metrics), which ushers a major change in nature of research
 - Also leads to automatic optimization to the evaluation criteria

Shared Tasks (1/2)

- Emergence of “shared tasks”
 - (Kaggle, WMT, SemEval, etc.)
- A task is **always** a formalized abstraction of a problem
- Examples:
 - User who will not look past the first 10 “hits” (precision-at-10 as metric)
 - Finding one relevant document with needed information (time-biased gain)
 - User who interacts with system (interactive IR evaluations)
- **Takeaway:** Don’t think in terms of “tasks.” Think about questions and problems.

Shared Tasks (2/2)



The “Leaderboard” Phenomenon and “SOTA-chasing”

- Hyper-attention to “tasks”, not problems or research questions
- Overfitting to specific datasets and formations
- Rank on leaderboard is not always equal to value or utility
- Incentive to focus only on innovation that steps a system up in the rankings using its single “figure of merit”
- Neglect of prediction cost (e.g. model size, training time, inference latency)
- Lack of robustness

How Evaluation Guides AI Research

- Cohen and Howe (*AI* magazine 1988) argue for evaluation to be integrated throughout the research cycle.
 1. Is the task significant? Why? Validate your formulation of the task.
 2. Is your research likely to meaningfully contribute to the problem?
Is the task tractable?
 3. As the task becomes specifically defined for your research, is it still representative of a class of tasks?
 4. Have any interesting aspects been abstracted away or simplified?
 5. How will you know when you have succeeded in demonstrating a solution?
Alternatively, is there a task where a solution can be demonstrated?

Evaluation Comparison: Elements of a “Results” Table

Baseline

lower bound, typically a straightforward approach

Previous A

Previous B

previous literature, existing state of the art

Ours #1

Ours #2

new approach(es)

Oracle or inter-annotator agreement

upper bound

Absolute vs. Relative Improvement vs. Error Reduction

Baseline: 80%

Ours: 85%

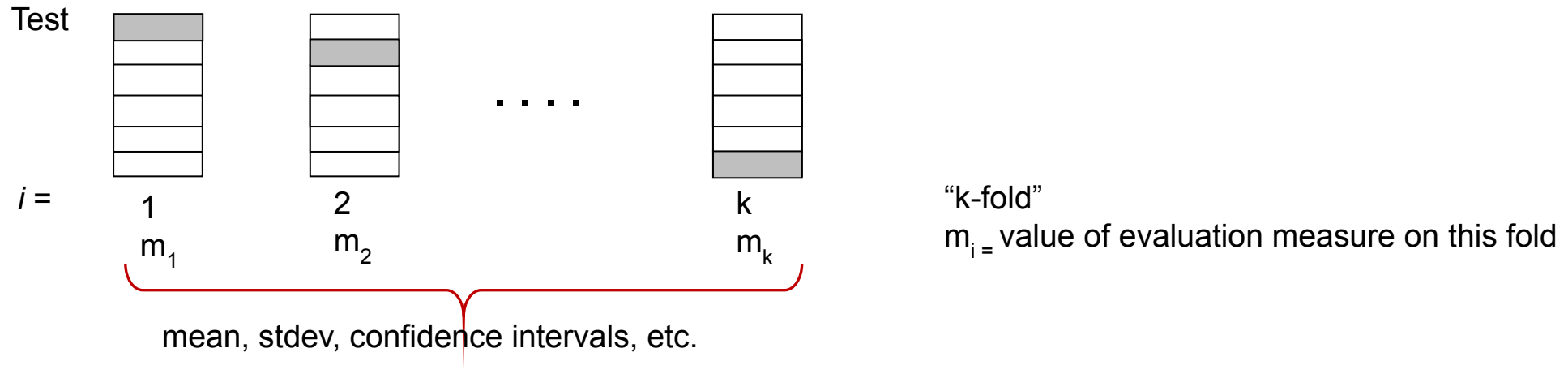
- Absolute improvement: $85\% - 80\% = 5\%$
- Relative improvement: $(85\% - 80\%) / 80\% = 6.25\%$
- Relative reduction in error rate: $(20\% - 15\%) / 20\% = 25\%$
- Notice: 98% to 99% is only 1% absolute, which seems unimpressive – but this is a full 50% reduction in the error rate

Partitioning of Data for Evaluation

- E.g. 70% for training, 20% for dev/tuning, 5% for devtest (formative), 5% for test (summative)
- Cardinal rule: don't test on training data if you want predictive value
- Beware “leakage” of training data into test data
 - Overfitting underestimates true error rate

Cross-validation

- Provides for intervals rather than point estimates of performance (cf. bootstrapping)
- Uses all the data while still properly maintaining train/test split



- Stratified cross-validation - preserves proportions of labels in each split

Inter-annotator Agreement and Upper Limits

- Cf. Chang *et al.*: “agreement” measures goodness
- Gale et al. (1992): human agreement can act as an upper bound on performance
- Accuracy = # Agreements / # Items
- But there is also some agreement by chance....

Chance-corrected agreement

$$\text{observed agreement} \rightarrow \frac{A_0 - A_e}{1 - A_e} \leftarrow \text{estimate of chance agreement}$$

Definitive reference: Artstein, R., & Poesio, M. (2008). Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4), 555-596.

Example Calculation of Cohen's Kappa

$$K = \frac{A_o - A_e}{1 - A_e}$$

A_o = observed agreement
 A_e = agreement by chance
 (if $A_e = 0$, denominator equals A_o)

Observed agreement:

$$A_o = \left(\frac{10 + 16 + 8}{50} \right) = 34/50 = 68\%$$

		Rater 1			
		psychotic	borderline	none	
Rater 2	psychotic	10	4	1	15
	borderline	6	16	2	24
	none	0	3	8	11
		16	23	11	(50 total)

diagonal counts observed agreements

Chance agreement **assuming independence of annotation**:

$$\frac{N_{\text{psychotic, rater1}}}{N} \frac{N_{\text{psychotic, rater2}}}{N} = \frac{16}{50} \times \frac{15}{50} = .32 \times .3 = .096$$

$$A_e = \sum_{k \in \{\text{psychotic, borderline, none}\}} \left(\frac{N_{k,1}}{N} \times \frac{N_{k,2}}{N} \right) = \frac{1}{N^2} \sum_k N_{k,1} N_{k,2}$$

Computation left as an exercise...

$K = 0.496$

Another Example Calculation of Cohen's Kappa

		J1		
		k=1	k=2	
J2	k=1	14	2	16
	k=2	2	22	24
		16	24	N = 40

$$A_0 = \frac{14 + 22}{40} = \frac{36}{40} = \frac{9}{10} = .9$$

$$A_{k=1} = \left(\frac{16}{40}\right) \left(\frac{16}{40}\right) \Rightarrow A_e = \frac{16^2}{40^2} + \frac{24^2}{40^2}$$

$$A_{k=2} = \left(\frac{24}{40}\right) \left(\frac{24}{40}\right) = \frac{256 + 576}{40^2} = \frac{832}{1600} = 0.52$$

$$K = \frac{A_o - A_e}{1 - A_e} = \frac{.9 - .52}{.48} = \frac{.38}{.48} = .792$$

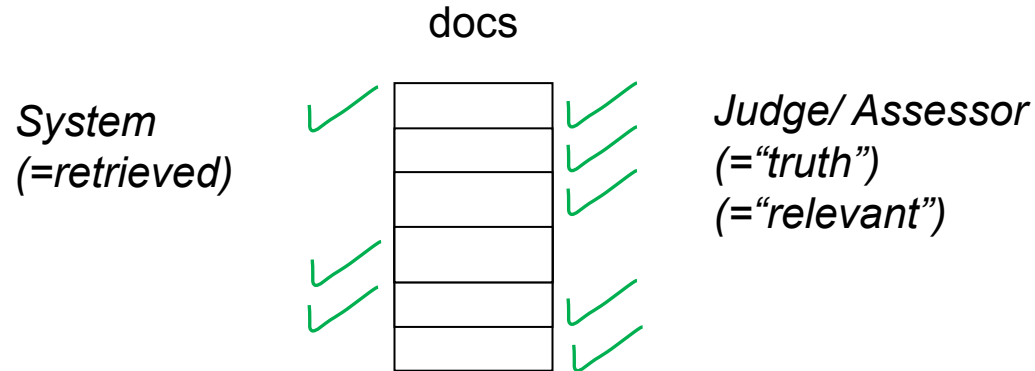
Chance-corrected agreement lower than $A_o = .9$
because of 0.52 probability of chance agreement

Evaluation with One Output Per Input, Exact Match

- Ex: Part-of-speech tagging
- Accuracy: number correct / number of items
- Variations:
 - Breakdown by category
 - Accuracy on just ambiguous cases (helps avoid inflated performance)
 - Allowing system to abstain from predictions:
 - **Accuracy** = # correct / # attempted
 - **Coverage** = # attempted / # items
 - **Recall** = # correct / # items
 - (Definitions vary, read carefully!)
 - Confusion matrix

Evaluation with Multiple Responses Per Item

- Ex: Document retrieval



Precision =
 $|\text{retrieved and relevant}| / |\text{retrieved}| = 2/3$

Recall =
 $|\text{retrieved and relevant}| / |\text{relevant}| = 2/5$

Consider extrema:

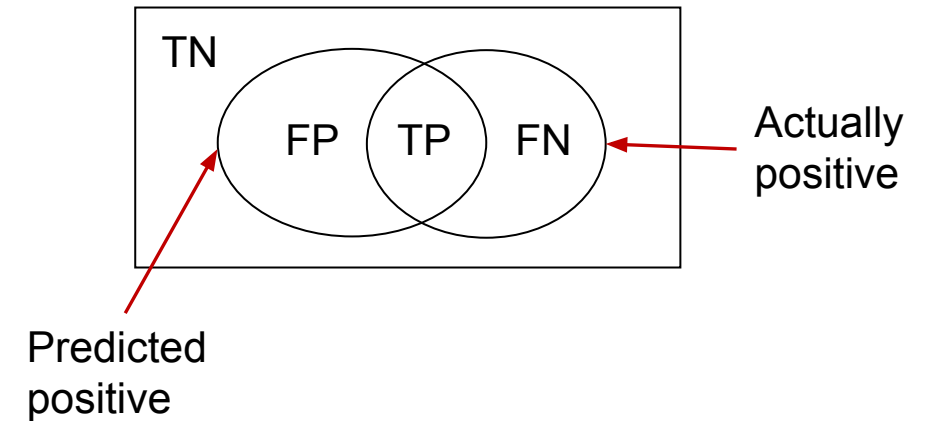
- Retrieve all documents: 100% recall!
- Retrieve no documents: max precision (no false positives!)

Evaluation with Multiple Responses Per Item

		Truth	
		+	-
Prediction	+	TP Type I error	FP Type I error
	-	FN "Miss" Type II error	22

Predicted positives

Actual positives



Combining Precision and Recall Into a Single Figure of Merit: F-measure

$$F = \frac{1}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}}$$

weighted
harmonic mean

$$\frac{2PR}{P+R}$$

where $\alpha = \frac{1}{2}$

common case,
equally weighting
precision and recall

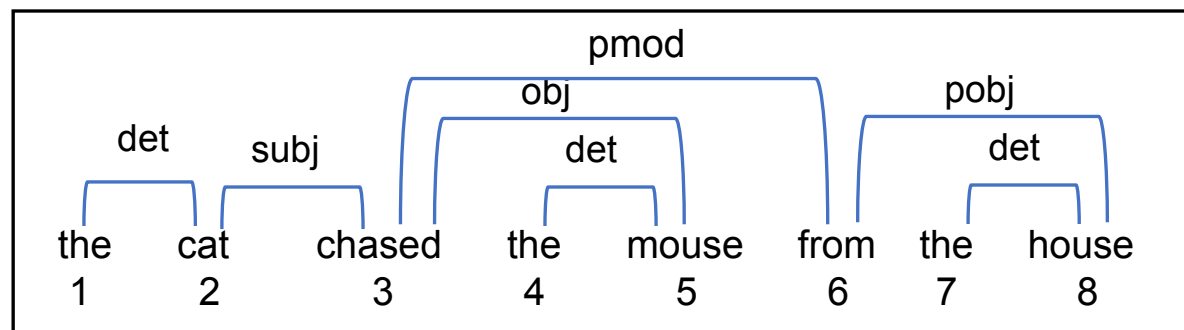
- Provides a single figure of merit for comparison/ranking
- Penalizes very different P/R values (hence discourages trading one off against the other)
- Biased toward maximizing true estimates, compared to accuracy's sensitivity to only the number of classification errors

Evaluating Structured Objects

Q: How do we evaluate output like syntactic dependency trees, which are structured?

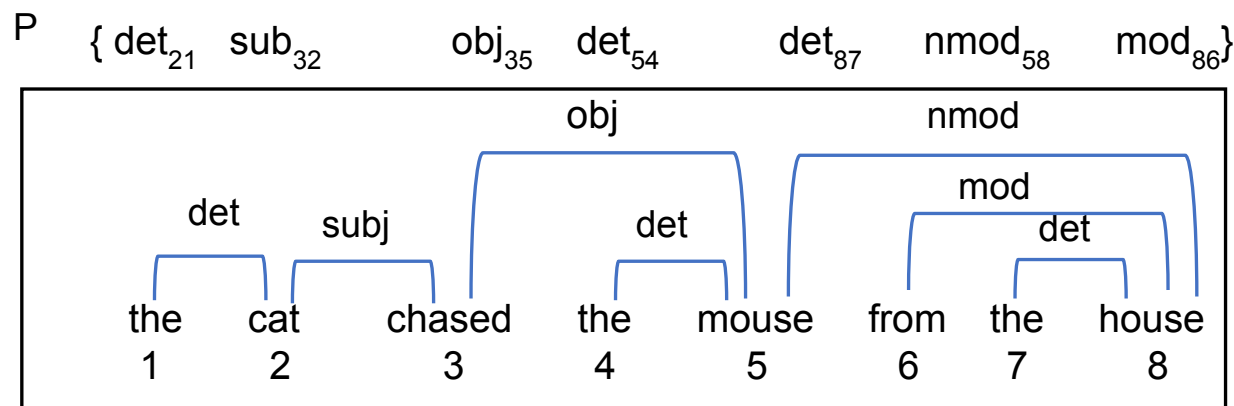
A: Break the structures into flat sets of items, and then use precision and recall.

True
parse



T { det₂₁ sub₃₂ pmod₃₄ obj₃₅ det₅₄ pobj₆₈ det₈₇ }

Predicted
parse



P { det₂₁ sub₃₂ obj₃₅ det₅₄ det₈₇ nmod₅₈ mod₈₆ }

true positives
($P \cap T$) = 5

false positives
($P - T$) = 2

false negatives
($T - P$) = 2

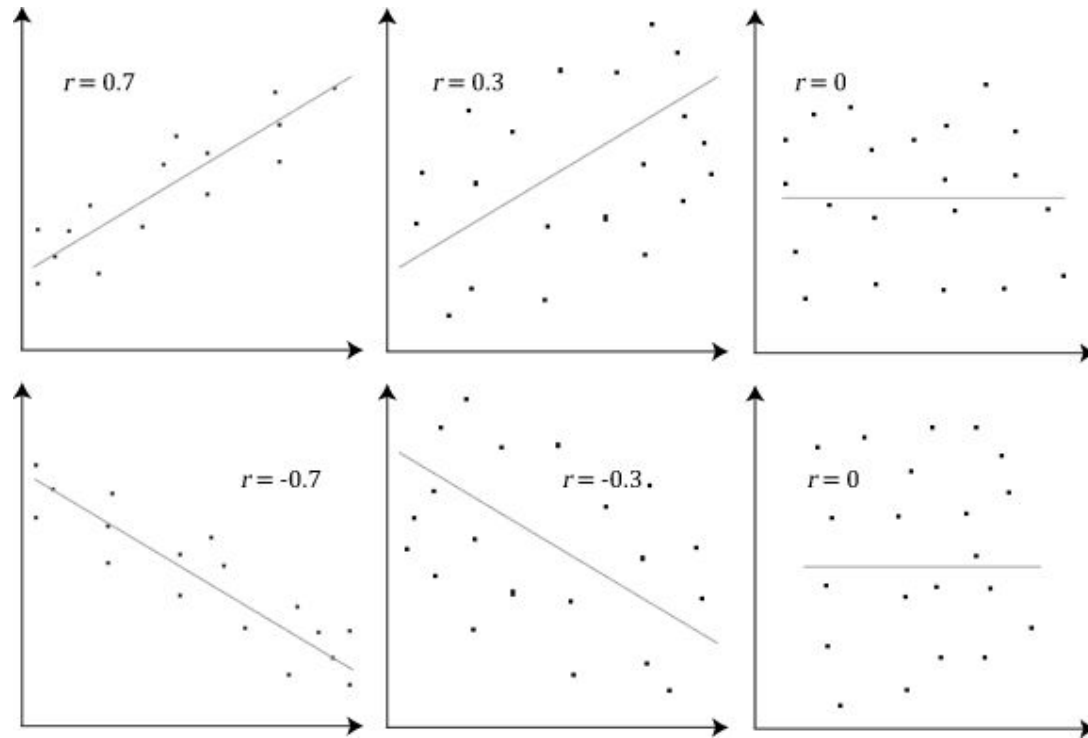
Precision
 $TP / (TP + FP) = 5/7$

Recall
 $TP / (TP + FN) = 5/7$

Evaluation Against Human Ratings

- What do human ratings look like?
 - **Discrete:**
 - In MT, assess **fluency** and **adequacy** using a scale (usually Likert-scale [5- or 7- point])
 - In sentiment classification, label positive/negative/neutral
 - **Continuous:** e.g. [0,1]
 - Strongly negative (0) to strongly positive (1) for sentiment polarity of a text
- Can be compiled from crowdsourcing platforms, e.g., Mechanical Turk

Evaluation Against Human Ratings



- **Question:** How do we evaluate with human ratings, especially when judging automatic metrics?
- **Typical answer:** Correlation between automatic and human assessments

Image Source: Laerd Statistics -
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commons.wikimedia.org/w/index.php?curid=84255641

Evaluation Against Human Ratings

- Issues to discuss
 - Need for data across the entire range
 - Interpretation of r^2 as % of variance
 - Multivariate regression